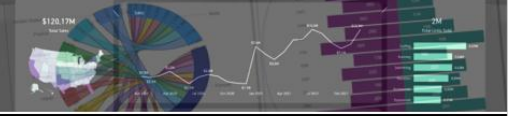




INTERMEDIATE

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SUMMARY

 IMPORTING THE DATA 	 BASIC DAX MEASURES 	 Creating the pages 	 CROSS-HIGHLIGHTING / FILTERING 
 Combo chart Combining lines & columns + average 	 BOOKMARKS 	 FIELD PARAMETER 	 Visual DAX 
 Calculation groups 	 DRILLTHROUGH PAGE 	 TOOLTIP PAGE 	 Introduction to DAX QUERY VIEW 

Queries [8]

- Medal Sort table
- Paris2024_ATHLETES
- Paris2024_MEDALS
- Paris2024_NOCS
- Paris2024_EVENTS
- dimDisciplines
- 01.Measures
- Z IMG

fx = Table.RenameColumns(#"Paris2024_ATHLETES développé",{"code", "AthleteID"})

medal_code	medal_date	A ^B name	1 ² NOCID	A ^B compound key	1 ² DisciplineID	A ^B Name in Athletes table	1 ² AthleteID
1	07.08.2024	Artur ALEKSANYAN	10	artur aleksanyan - 10		316 ALEKSANYAN Artur	1532872
2	07.08.2024	Malkhas AMOYAN	10	malkhas amoyan - 10		314 AMOYAN Malkhas	1532873
3	11.08.2024	Tatiana RENTERIA REN...	48	tatiana renteria renteria...		323 RENTERIA RENTERIA Tatia...	1535513
4	09.08.2024	Shanieka RICKETTS	106	shanieka ricketts - 106		59 RICKETTS Shanieka	1536888
5	26.07.2024	Federico Nilo MALDINI	104	federico nilo maldini - 1...		217 MALDINI Federico Nilo	1551386
6	28.07.2024	Paolo MONNA	104	paolo monna - 104		217 MONNA Paolo	1551387
7	09.08.2024	Marco Alonso VERDE A...	133	marco alonso verde alvare...		84 VERDE ALVAREZ Marco Alo...	1535187
8	30.07.2024	Daniel WIFFEN	99	daniel wiffen - 99		261 WIFFEN Daniel	1539969
9	04.08.2024	Daniel WIFFEN	99	daniel wiffen - 99		262 WIFFEN Daniel	1539969
10	08.08.2024	Osmar OLVERA IBARRA	133	osmar olvera ibarra - 133		138 OLVERA IBARRA Osmar	1535345

Query Settings

PROPERTIES

Name

Paris2024_MEDALS

All Properties

APPLIED STEPS

Source

En-têtes promus

Type modifié

Autres colonnes supprimées

Colonnes permutées

IMPORTING THE DATA



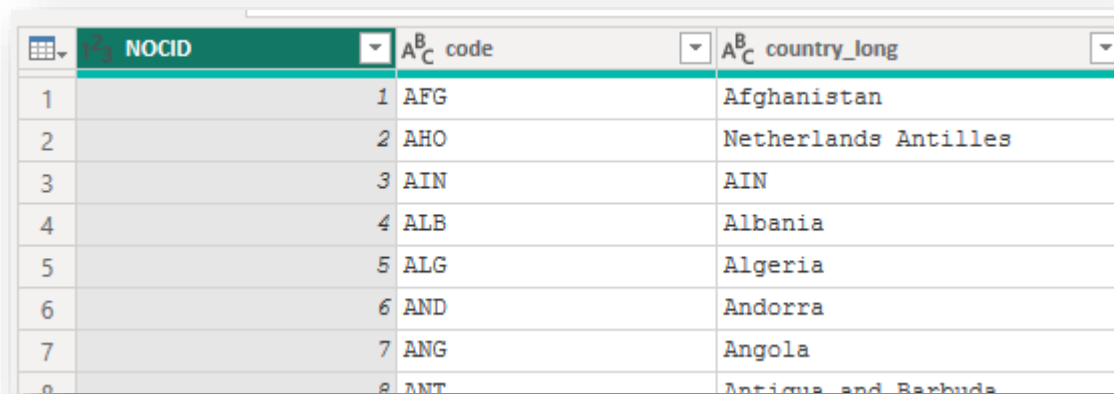
Power BI
Desktop

Connecting to data sources and cleanup

Import the nocs.csv (CSV file) from Github with CSV connector of Power Query, the URL being:

<https://github.com/artofnet-training/PBI-INT/raw/refs/heads/main/Paris2024/nocs.csv>

- Promote headers.
- Keep only these columns: [code],[country_long].
- Add an index column, rename it [NOCID] and place it in 1st position.
- Check the column types.



A screenshot of a Power Query table. The table has three columns: 'NOCID', 'code', and 'country_long'. The 'NOCID' column is the first column and contains an index from 1 to 8. The 'code' column contains three-letter country codes. The 'country_long' column contains the full names of the countries. The table is displayed in a grid format with a light blue header and alternating light blue and white rows.

	NOCID	code	country_long
1		AFG	Afghanistan
2		AHO	Netherlands Antilles
3		AIN	AIN
4		ALB	Albania
5		ALG	Algeria
6		AND	Andorra
7		ANG	Angola
8		ANT	Antigua and Barbuda

Connecting to data sources and cleanup

Import the events.csv (CSV file) from Github with CSV connector of Power Query, the URL being:

<https://github.com/artofnet-training/PBI-INT/raw/refs/heads/main/Paris2024/events.csv>

- Promote headers.
- Keep only these columns: [event], [sport] and [sport_code].
- Add an index column, rename it [EventID] and place it in 1st position.
- Check the column types.

EventID	event	sport	sport_code
Valid 100%	Valid 100%	Valid 100%	Valid 100%
Error 0%	Error 0%	Error 0%	Error 0%
Empty 0%	Empty 0%	Empty 0%	Empty 0%
1	Men's Individual	Archery	ARC
2	Women's Individual	Archery	ARC
3	Men's Team	Archery	ARC
4	Women's Team	Archery	ARC
5	Mixed Team	Archery	ARC
6	Men's Team	Archery	ARC

Connecting to data sources and cleanup

Import the athletes.csv (CSV file) from Github with CSV connector of Power Query, the URL being:

<https://github.com/artofnet-training/PBI-INT/raw/refs/heads/main/Paris2024/athletes.csv>

- Promote headers.
- Keep only these columns: [code],[name0_tv],[gender],[country_code].
- Merge the table with the NOCS table to insert the [NOCID] field on each row.
- Remove [country_code].
- Check the columns types.

code	name_tv	gender	NOCID
Valid 100%	Valid 100%	Valid 100%	Valid 100%
Error 0%	Error 0%	Error 0%	Error 0%
Empty 0%	Empty 0%	Empty 0%	Empty 0%
1532872	Artur ALEKSANYAN	Male	10
1532873	Malkhas AMOYAN	Male	10
1532874	Slavik GALSTYAN	Male	10
1532944	Arsen HARUTYUNYAN	Male	10
1532945	Vazgen TEVANYAN	Male	10

Connecting to data sources and cleanup

Import the medals.csv (CSV file) from Github with CSV connector of Power Query, the URL being:

<https://github.com/artofnet-training/PBI-INT/raw/refs/heads/main/Paris2024/medals.csv>

- Promote headers.
- Keep only these columns:
[medal_code],[medal_date],[name],[discipline],[event],[country_code].
- Merge the table with the Athletes table to insert the [code] field on each row and delete [country_code]. Use [name] and [name_tv] for merging.
- Merge the resulting table with the Events table to add the [EventID] key column. Use both [discipline] and [event] columns as the key columns for merging.
- Check the columns types.

Connecting to data sources and cleanup

Import the medallists.csv (CSV file) from Github with CSV connector of Power Query, the URL being:

<https://github.com/artofnet-training/PBI-INT/raw/refs/heads/main/Paris2024/medals.csv>

- Promote headers and filter [is_medallist] to True.
- Keep only these columns:
[medal_code],[medal_date],[gender],[discipline],[event],[birth_date] & [code_athlete].
- Merge the table with the Events table to insert the [EventID] field on each row and delete [discipline] and [event].
- Add a new column that calculates the age (in days) by subtracting the birth date from today's date.
- Check the columns types.

Creating helping tables

1. In the Home tab, click on 'Enter data' then give the new empty table a name (01.Measures for example) and click 'OK'.
This table will be used to store all the DAX measures needed for this report.
2. Create another manual table that will store the IMG URL that will be included in your pages. Name it 'Z.IMG' for example.
Add 2 columns [ID] and [URL] and fill the first row with the official Paris 2024 Logo picture. Repeat for the LA2028 logo and whatever picture you wish to include in your report.
3. Insert a new table and call it 'Medal Sort Table'. Insert a column for the type of Medal, another one for the Sort order and a third one that contains the medal_code info matching the values in the Paris2024_MEDALS table.

Create the data model

Model View

1. In the model view, check the detected relationships and add or delete as needed.
2. Add a Calendar table by inserting the DAX script provided for this purpose.
Name it 'CALENDAR'.
Configure the start and end dates in the DAX code.
Mark as a date table and select the date field in the calendar table as the date reference.
3. Create a 'hr_date' hierarchy in the CALENDAR table and add [Year], [Month], [Week Number], and [Date].
Don't forget to sort [Week Number] by the [Week] column.

```
DEFINE
MEASURE Sales[Same period last month] =
    CALCULATE (
        [Sales Amount],
        DATEADD ( 'Date'[Date], -1, MONTH )
    )
MEASURE Sales[Same period last quarter] =
    CALCULATE (
        [Sales Amount],
        DATEADD ( 'Date'[Date], -1, QUARTER )
    )
MEASURE Sales[Same period last year] =
```

```
EVALUATE
VAR StartDate = DATE ( 2008, 07, 25 )
VAR EndDate = DATE ( 2008, 07, 31 )
RETURN
    CALCULATETABLE (
        DATEADD ( 'Date'[Date], -1, YEAR ),
        'Date'[Date] >= StartDate &&
        'Date'[Date] <= EndDate
    )
ORDER BY [Date]
```

BASIC DAX MEASURES



Power BI
Desktop

DAX MEASURES

Create basic explicit DAX measures such as:

NB Medals = COUNTROWS(Paris2024_MEDALS)

NB GOLD = CALCULATE(COUNTROWS(Paris2024_MEDALS),
Paris2024_MEDALS[medal_code]=1
)

NB SILVER = CALCULATE(COUNTROWS(Paris2024_MEDALS),
Paris2024_MEDALS[medal_code]=2
)

NB BRONZE = CALCULATE(COUNTROWS(Paris2024_MEDALS),
Paris2024_MEDALS[medal_code]=3
)

NB days = DISTINCTCOUNT(Paris2024_MEDALS[medal_date])

DAX MEASURES

Create basic explicit DAX measures calculating the overall average number of medals:

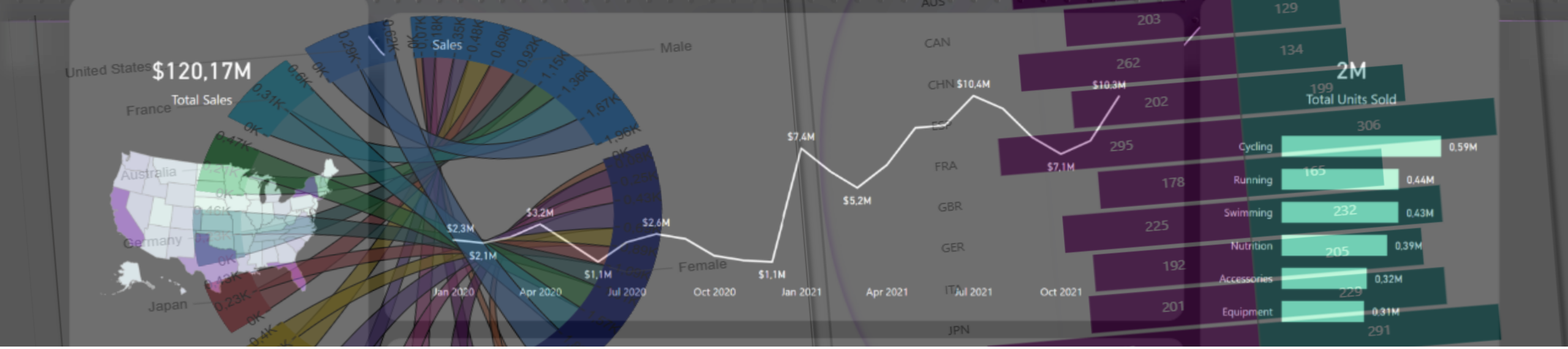
AVG medals per day (global) =

```
VAR __totdays = CALCULATE([NB days],  
                           REMOVEFILTERS(Paris2024_MEDALS))
```

```
VAR __totmedals = CALCULATE([NB Medals],  
                           REMOVEFILTERS(Paris2024_MEDALS))
```

```
RETURN
```

```
ROUNDDOWN(DIVIDE(__totmedals,__totdays),0)
```



Creating the pages



Power BI
Desktop

Start Page

- PURPOSE: To provide a start page with a navigation menu to all pages in the report.
- Format the background of page 1 with a Scrim (BigSquare.png).
Page Size>Canvas Background> Image
- Scrims available here:
<https://github.com/artofnet-training/PBI-INT/SCRIMS/SCRIMS-BLUE.zip>
- Download Markus Kolp's 'Image' visual and insert it into the page. In "Image URL", drag Z IMG[URL] and apply a filter on this visual on the Z IMG[ID] field in order to select the Paris2024 logo to be displayed.
- Insert a page navigation menu. Insert > Button > Browser > Page Browser.
- Insert a card visual displaying the remaining days till LA2028 (create a DAX measure) and insert the LA2028 logo in format > Image > image URL.
The Dax measure to compute the remaining days:
Countdown LA2028 = `INT( TODAY()-DATE( 2028, 7, 14)) & " days"`

The Athletes Page

1. Create three measures calculating the total number of athletes and the number of male/female athletes.
NB athletes = `COUNTROWS(Paris2024_ATHLETES)`
NB Men = `CALCULATE(COUNTROWS(Paris2024_ATHLETES), Paris2024_ATHLETES[gender]="Male")`
NB Women = `CALCULATE(COUNTROWS(Paris2024_ATHLETES), Paris2024_ATHLETES[gender]="Female")`
2. Insert these 3 measures into cards at the top of the page.
3. Insert a matrix with Paris2024_NOCS[country_long] as "Lines" and [NB athletes] as "Values".
4. Add conditional formatting for the values.
Format > Cell elements > Series: NB Athletes > Data Bars(turn on)
5. If you can, download a visual from the store. To do this, click the three dots 'Get more visuals' at the end of the list of visuals in the 'Build' panel.
Once in the store, search for Microsoft's 'tornado chart' and download.
6. Insert a Tornado visual in the page with NOCS[code] in "Group", Paris2024_ATHLETES[gender] in "Legend" and [NB athletes] in "Values".
7. Add a split screen scrim for background

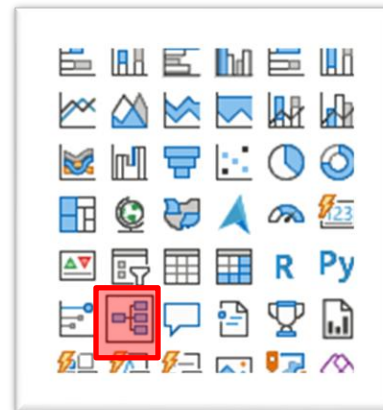
The 'Athletes' page



Create a decomposition tree

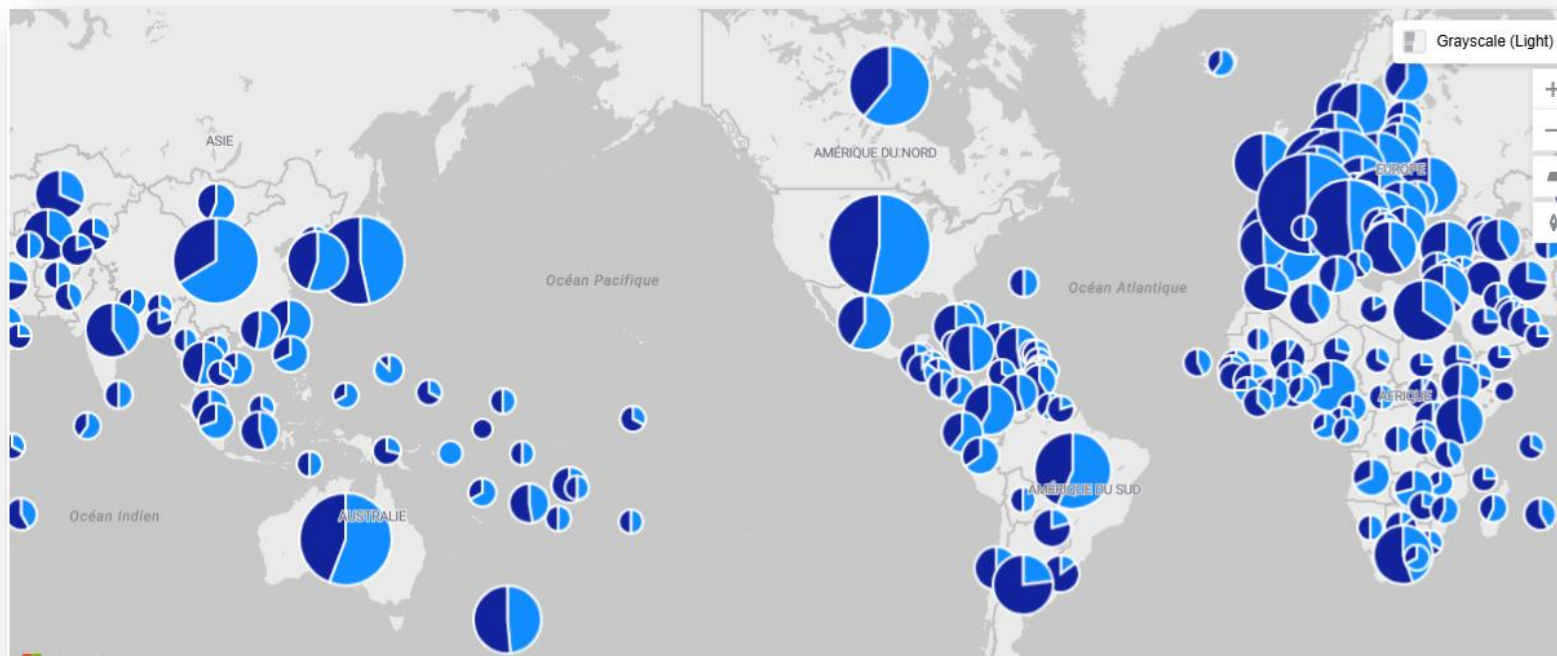
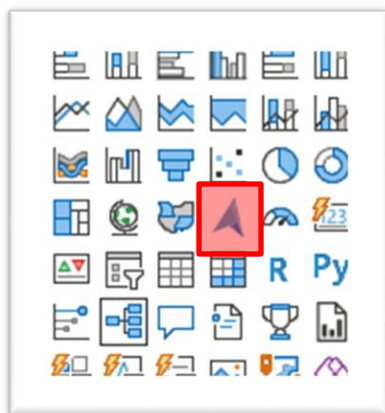
In a new page, insert a 'decomposition tree' visual and drag [NB Medals] into the 'Analyze' well.

Drag Medal Sort Table[Medal],
Paris2024_NOCS[country_long],
Paris2024_EVENTS[sport], and
Paris2024_EVENTS[event] into "Explain by."



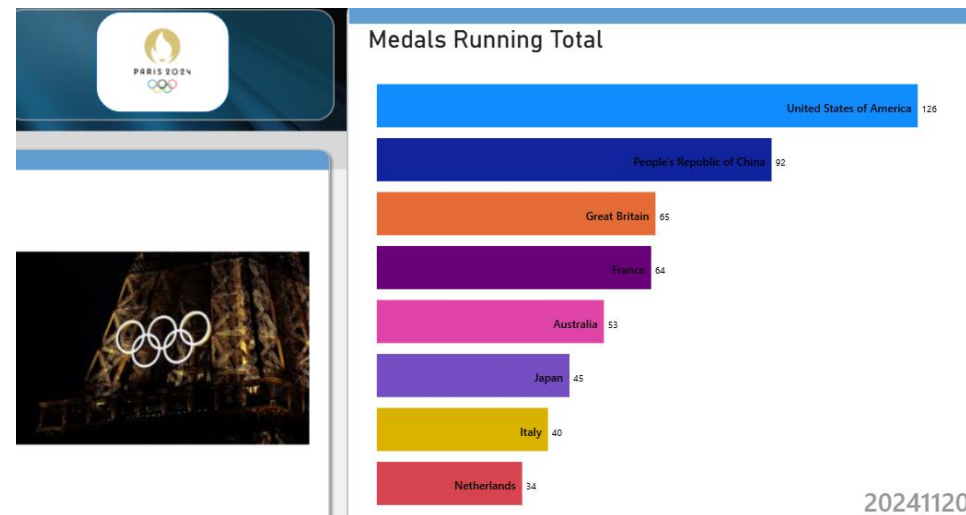
Create an Azure Maps visual

Insert an Azure Maps visual and drag Paris2024_NOCS[country_long] into "Location", [NB athletes] into "Size" and Paris2024_ATHLETES[gender] into "Legend".



Animated bar chart

- Download a new 'animated bar chart race' (designed by Wishyoulization) from the store and insert it in a new page.
- Create a DAX measure that computes the Running sum of medals per day:
Running Total Medals = **CALCULATE**(
 [NB Medals],
 'CALENDAR'[Date] <= **MAX**('CALENDAR'[Date]))
- Add Paris2024_NOCS[country_long] in "Name", [Running Total Medals] in "Value" and Calendar[Date as integer] in "Period" (period must be numeric for display).
- Go to Format > Visual Object > Title and insert the title of your choice.
- Go to Format > Visual Object > Wishyoulization Settings > Duration and insert 2000 or more, enable the 'hideGrid' option and enter 32 in 'periodSize'. If necessary, change the value of 'TopN' to display fewer results during the animation.
- Double click on the visual to start the animation.



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Insert a Row/Column Combo Visual

- Insert a combo visual with [NB Medals] in "Line Y-Axis" and [Date] in "X-Axis".
- Create a metric that calculates daily average of medals and add it to "Y Axis of Line":

AVG medals =

```
VAR __NBmedals = CALCULATE([NB Medals],ALLSELECTED('CALENDAR'[Date]))
```

```
VAR __NBdays = CALCULATE([NB days],REMOVEFILTERS(Paris2024_MEDALS))
```

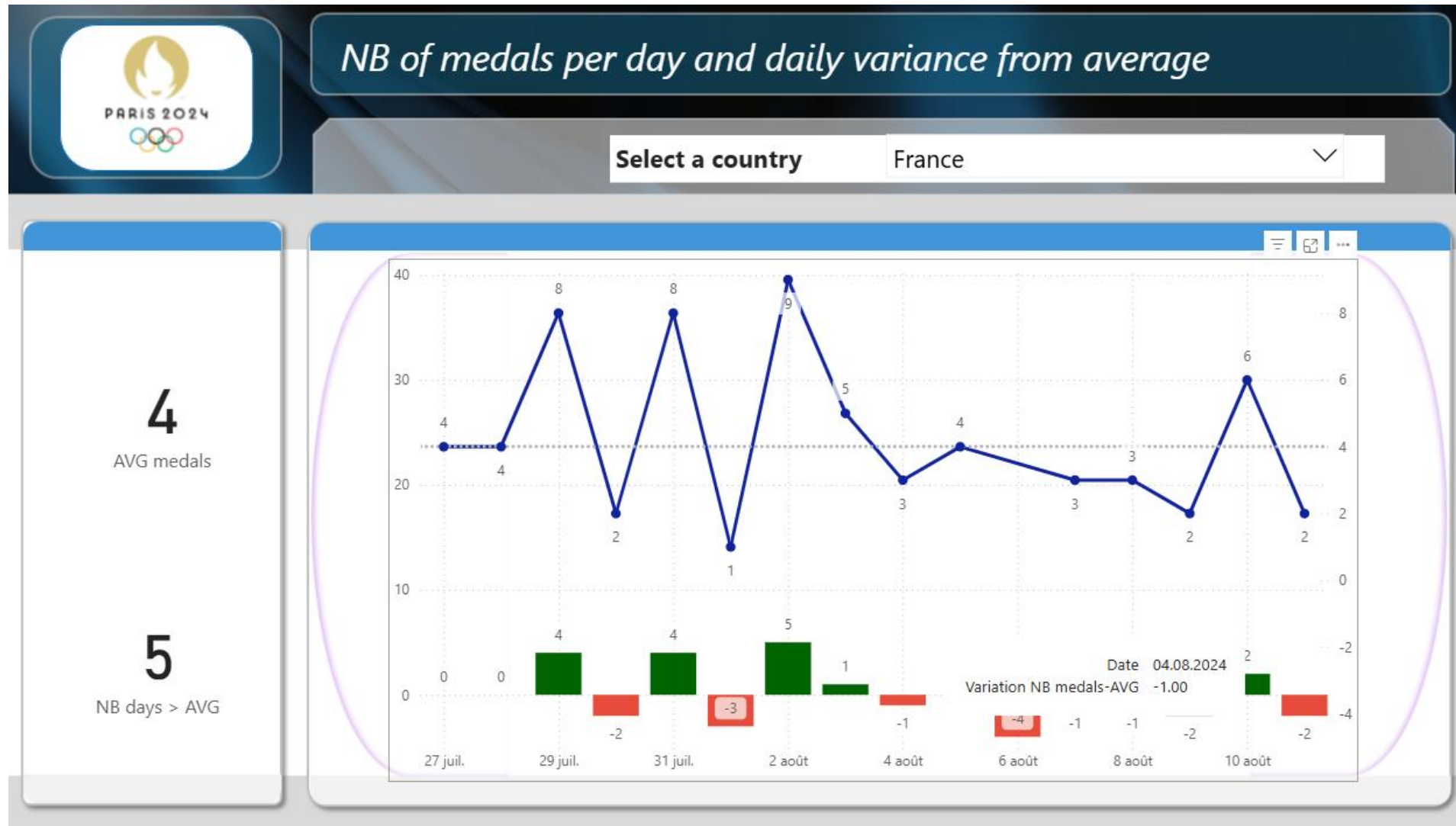
```
VAR __result = DIVIDE(__NBmedals,__NBdays)
```

```
RETURN
```

```
ROUNDDOWN(__result,0)
```

- Create a 2nd measure that calculates the variation between the number of medals and the average:
Variation NB medals-AVG = [NB Medals] - [AVG medals]
Add it to "Y-axis of the column".
- Format the axes to clearly visualize the three measurements.
- Add a segment (drop-down list) on the country and a visual map that displays [AVG Medals].
- Add a segment (drop-down list) on Medal Sort Table[Medal]

The 'AVG-Variation' page



Ribbon chart with age ranges

- Create the following DAX metric:
NB Medallists = `COUNTROWS`(Paris2024_MEDALLISTS)
- Add a calculated column in Paris_2024_MEDALLISTS:
Age (y) = `INT`(`DIVIDE`(Paris_2024_MEDALLISTS[Age],365.25))
- Right-click on Age (y) in the 'Data' panel and select 'New Group'.
- Choose a bin size of 5

Groups [X]

Name *	Field
Age(ans) (compartiments) (bins)	Age(ans) (compartiments)
Group type	Bin type
Bin	Size of bins
Min value	Max value
15	60

Binning splits numeric or date/time data into equally sized groups. Enter bin size.

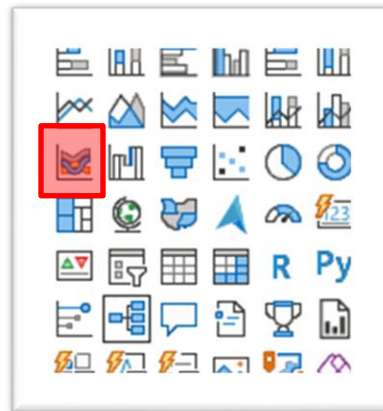
Bin size *

5

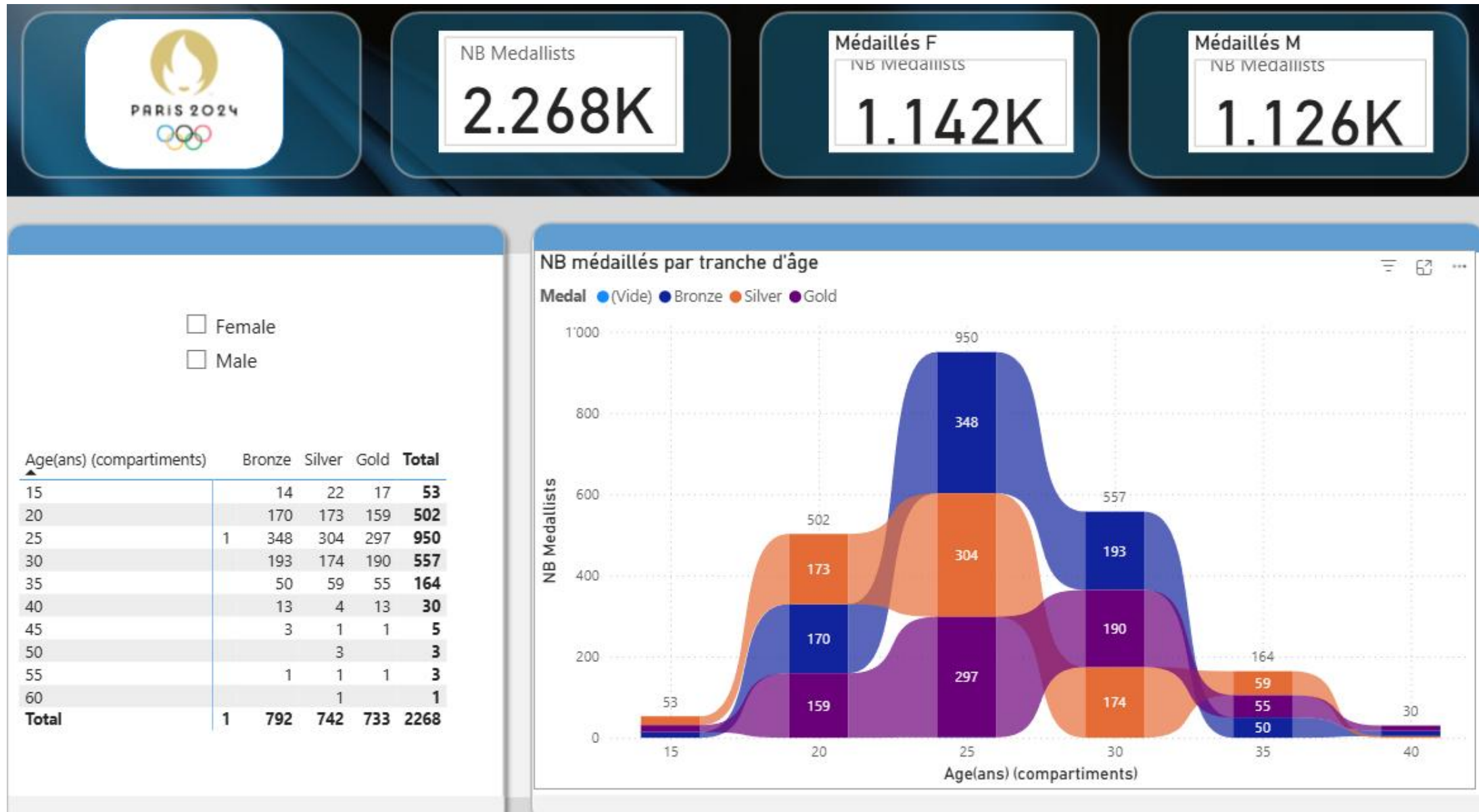
Reset to default

Ribbon chart

- Insert a ribbon chart with [age (y)] in "X-axis", [NB Medallists] in "Y-axis" and Medal Sort table[Medal] in "Legend". View Data Labels (Format)Visual Object > Data Labels and adjust the density in the same option group as desired. Change the title.
- Filter the ribbon chart on [age(y) (bins)] and keep values between 15 and 40.
- Insert a matrix visual with [age (y) (bins)] in "Rows", [NB Medallists] in "Values" and Medal Sort table [Medal] in "Columns".
- Insert a segment on Paris2024_ATHLETES[gender].
- Insert 3 cards that display [NB Medallists] as the grand total + number of F and M medallists.
- Disconnect the slicer from these 3 total cards.



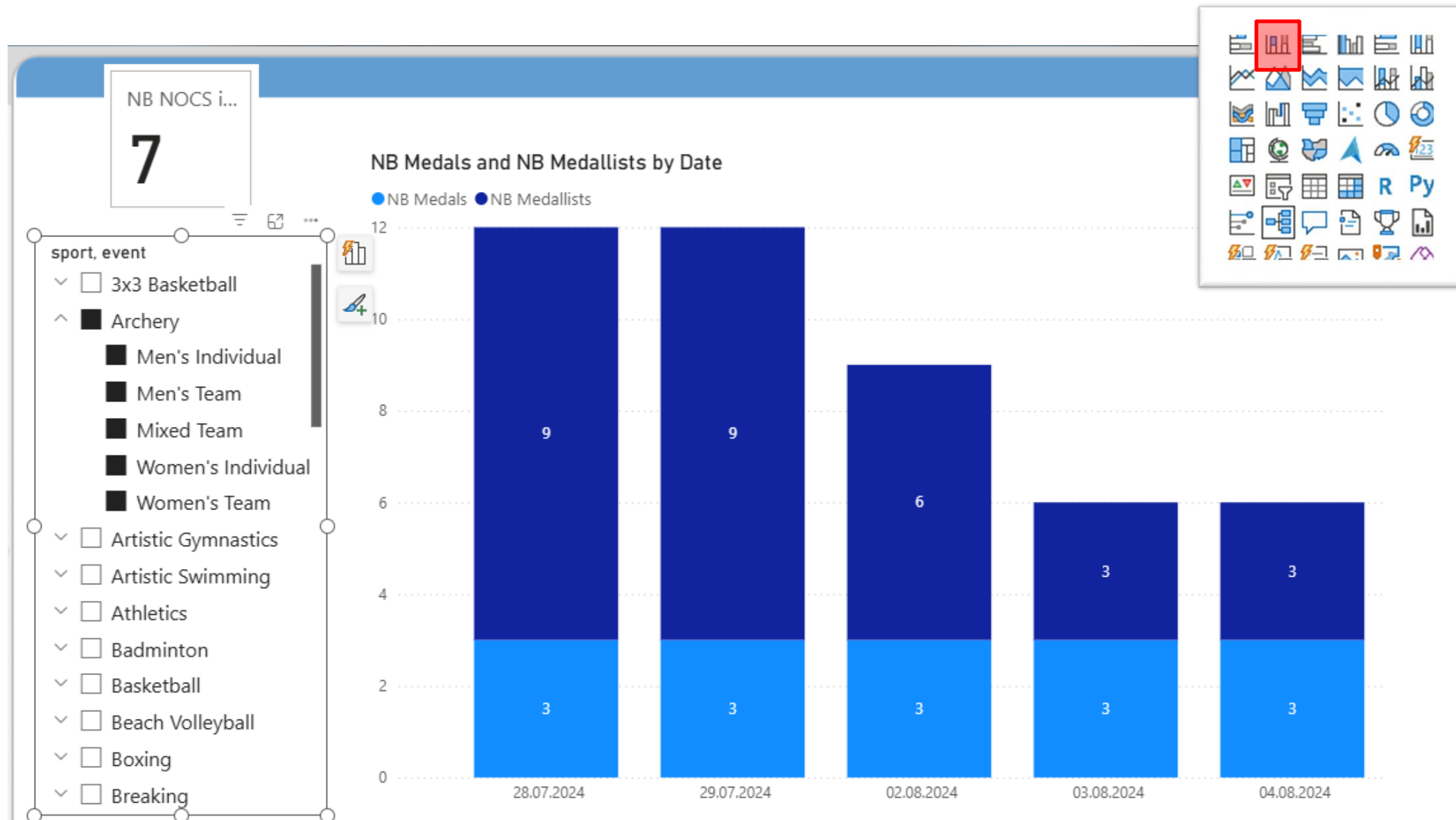
The 'Medallists' page



Stacked Histogram

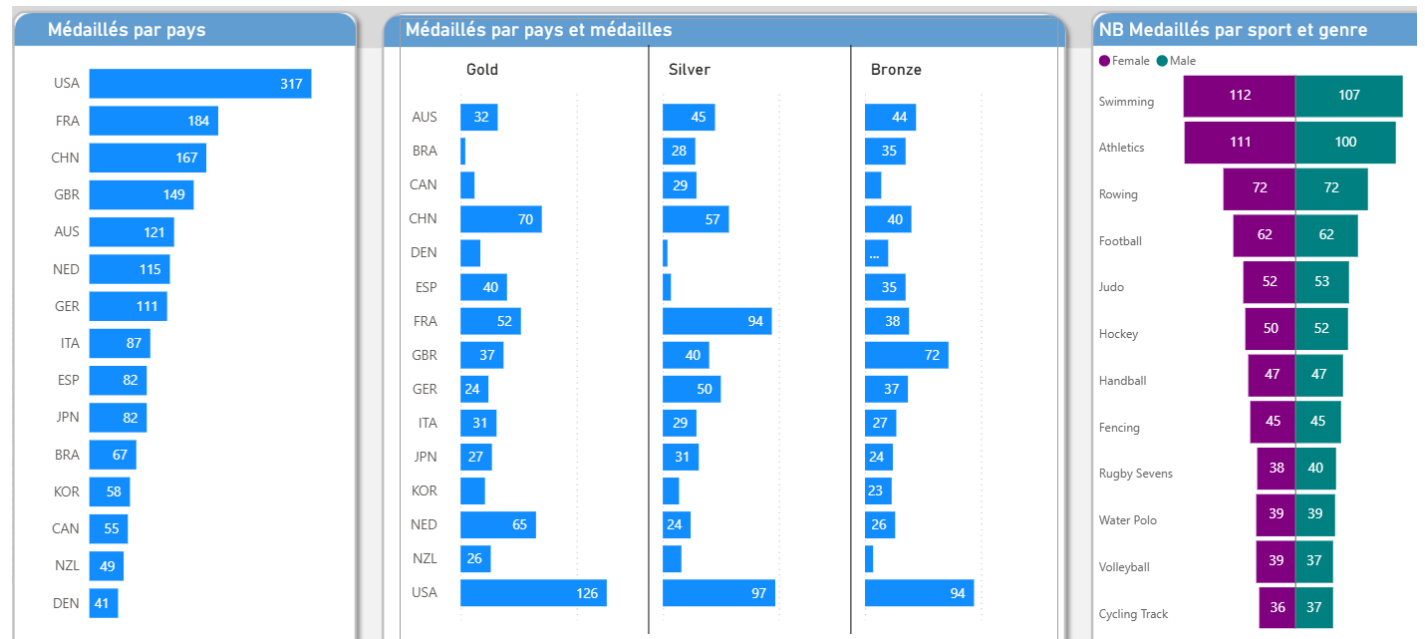
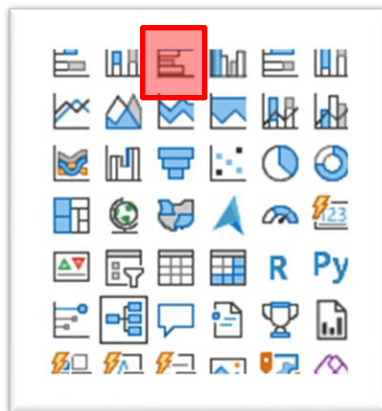
- Insert a stacked histogram with Calendar[Date] in "X-Axis" and [NB Medals] + [NB Medallists] in "Y-Axis".
- Enable data labels and adjust the density (Format > Data Labels > Options > Label Density).
- Insert a slicer and drag Paris2024_EVENTS[sport] and Paris2024_EVENTS[event] to the Field well.
- Format the X-axis type to categorical to avoid automatic date/time conversion.
- Create and display in a card visual a DAX measure that calculates the number of countries involved in the chart:
NB NOCS in Medals = **DISTINCTCOUNT**(Paris2024_MEDALS[NOCID])
- Explain why COUNTROWS(Paris2024_NOCS[NOCID]) does not yield the same result.

Stacked Histogram



Small multiples and tornado chart

- Insert a grouped bar design with Paris2024_NOCS[code] in "Y-Axis" and [NB Medallists] in "X-Axis".
- Duplicate this visual and add Medal Sort Table[Medal] in "Small Multiples", format in 1 row and 3 columns.
- Add a Tornado visual with Paris2024_EVENTS[sport] in "Group", Paris2024_MEDALLISTS[gender] in "Legend" and [NB Medallists] in "Values" on the right.





CROSS-HIGHLIGHTING / FILTERING



Power BI
Desktop

CROSS-HIGHLIGHTING / CROSS-FILTERING

In Power BI, when you click on or select part of a visual (like a bar, line, or point), that interaction can affect the other visuals on the same page in one of two ways:

1. **Cross-filtering** → means that selecting an element in one visual *filters* the data shown in other visuals so that only related records remain visible. Common in visuals like **slicers**, **tables**, and **cards**.
2. **Cross-highlighting** → dims unrelated data in other visuals but keeps all values visible, highlighting only the relevant ones. Works best with **bar**, **column**, or **pie** visuals.

You can define whether a visual **cross-filters** or **cross-highlights** others:

Select a visual → Go to the **Format ribbon** → **Edit interactions**

On each other visual, you'll see icons:

Filter icon → Cross-filter

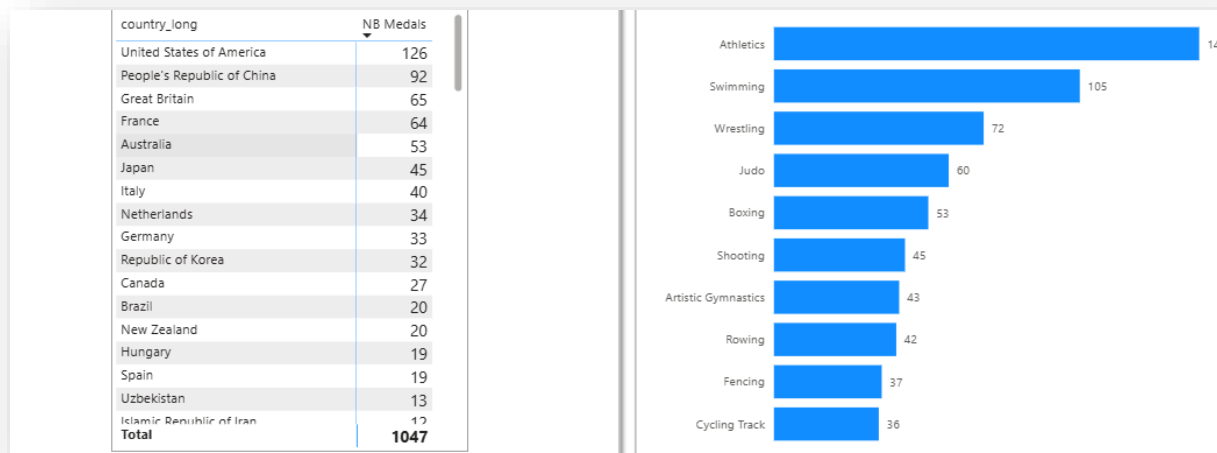
Highlight icon → Cross-highlight

No icon → No effect

CROSS-HIGHLIGHTING / CROSS-FILTERING

1. Insert a table visual with Paris2024_NOCS[country_long] and [NB medals] in the rows well.
2. Insert a bar chart with dimDisciplines[sport] in the Y-axis well and [NB_medals] in the X-axis well.
3. Duplicate the bar chart.
4. Select the table then edit the interactions (Format > Edit interactions) and click on the filter button on the bar chart.

Click on one of the countries in the table visual to see it cross-highlighting the 1st bar chart and cross-filtering the 2nd one.





Combo chart

Combining lines & columns + average



Power BI

Desktop

Combo chart exercise : objective

- **GOAL:**

Create a line chart that displays the number of medals per day. You have to add a line for the global average of medals over all days.

- Add a column chart that will display the daily variation from the global average value.
- There should be a slicer with the list of countries in the header section that filters the results for the selected country.
- You have to add two cards displaying the average number of medals and the number of days where the variation was positive (NB medals > average).

Combo chart exercise : Solution

- Create the DAX measure that calculates the average number of medals over all the days:

AVG medals =

```
VAR __NBmedals = CALCULATE([NB Medals],ALLSELECTED('CALENDAR'[Date]))
VAR __NBdays = CALCULATE([NB days],REMOVEFILTERS(Paris2024_MEDALS))
VAR __result = DIVIDE(__NBmedals,__NBdays)
RETURN
ROUNDDOWN(__result,0)
```

- Create a measure that calculates the number of days where the number of medals was greater than the average.

NB days > AVG =

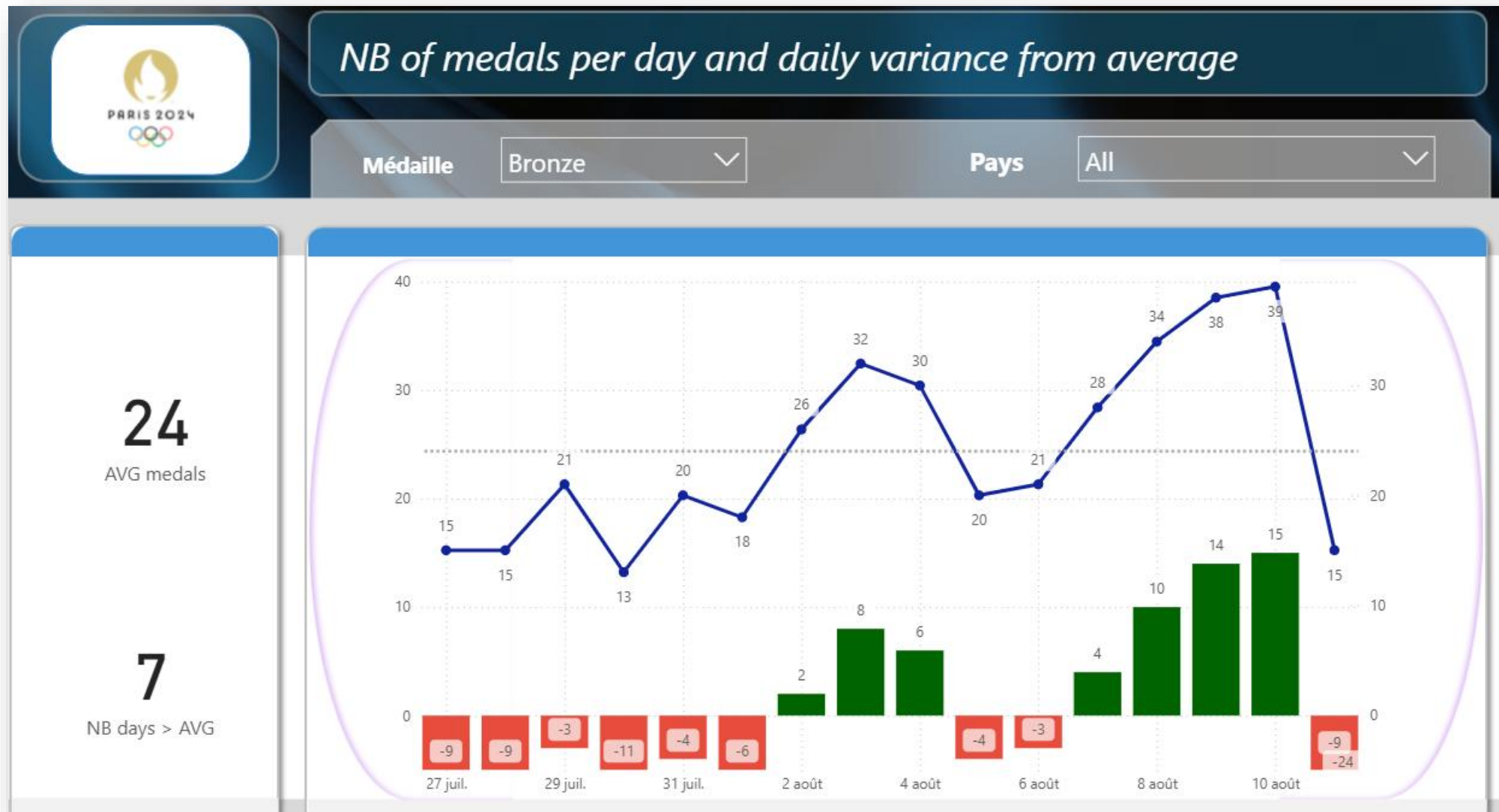
```
VAR __threshold = [AVG medals]
VAR __vszl = SUMMARIZE(Paris2024_MEDALS,'CALENDAR'[Date],"@NB",[NB Medals])
VAR __vfl = FILTER(__vszl, [@NB] > __threshold)
VAR __result = COUNTROWS(__vfl)
RETURN __result
```

- Create a measure that calculates the difference:
Variation NB medals-AVG = [NB Medals] - [AVG medals]

Combo chart exercise : Solution

- Insert a combo chart visual with CALENDAR[Date]. Drag [NB medals] and [AVG medals] in the Line y-axis well and [Variation NB medals-AVG] in the Column y-axis.
- Turn the data labels on and create a DAX measure to apply conditional formatting on the columns colors (red when negative and green when positive).
Color for variance = `IF([Variation NB medals-AVG] >=0, "darkgreen", "#E74C3C")`
- Modify the Y-axis and secondary y-axis min and max values so that the lines and the columns don't cross.
- Add 2 card visuals to display [AVG medals] and [NB days > AVG].
- Insert a scrim background and a slicer with the list of countries.

Combo chart





BOOKMARKS



Power BI
Desktop

Power BI – What is a bookmark ?

In Power BI, a **bookmark** is a saved *snapshot* of a report page in its current state — including filters, slicer selections and visuals' visibility,

Typical uses:

- **Interactive navigation** – Create buttons that let users move between filtered views (e.g., “Show Top 10”, “Show All Countries”).
- **Storytelling / Presentations** – walk through insights step-by-step.
- **Custom views** – Allow users to switch between visuals or layouts (e.g., “Map View” vs. “Table View”).

Process:

Create several snapshots (bookmarks) of a report page then add a bookmark navigator (Insert > Buttons > bookmark navigator) to display the bookmark of your choice.

Bookmarks: Exercise

Create 3 DAX measures for calculating the number of Gold, silver and bronze medals.

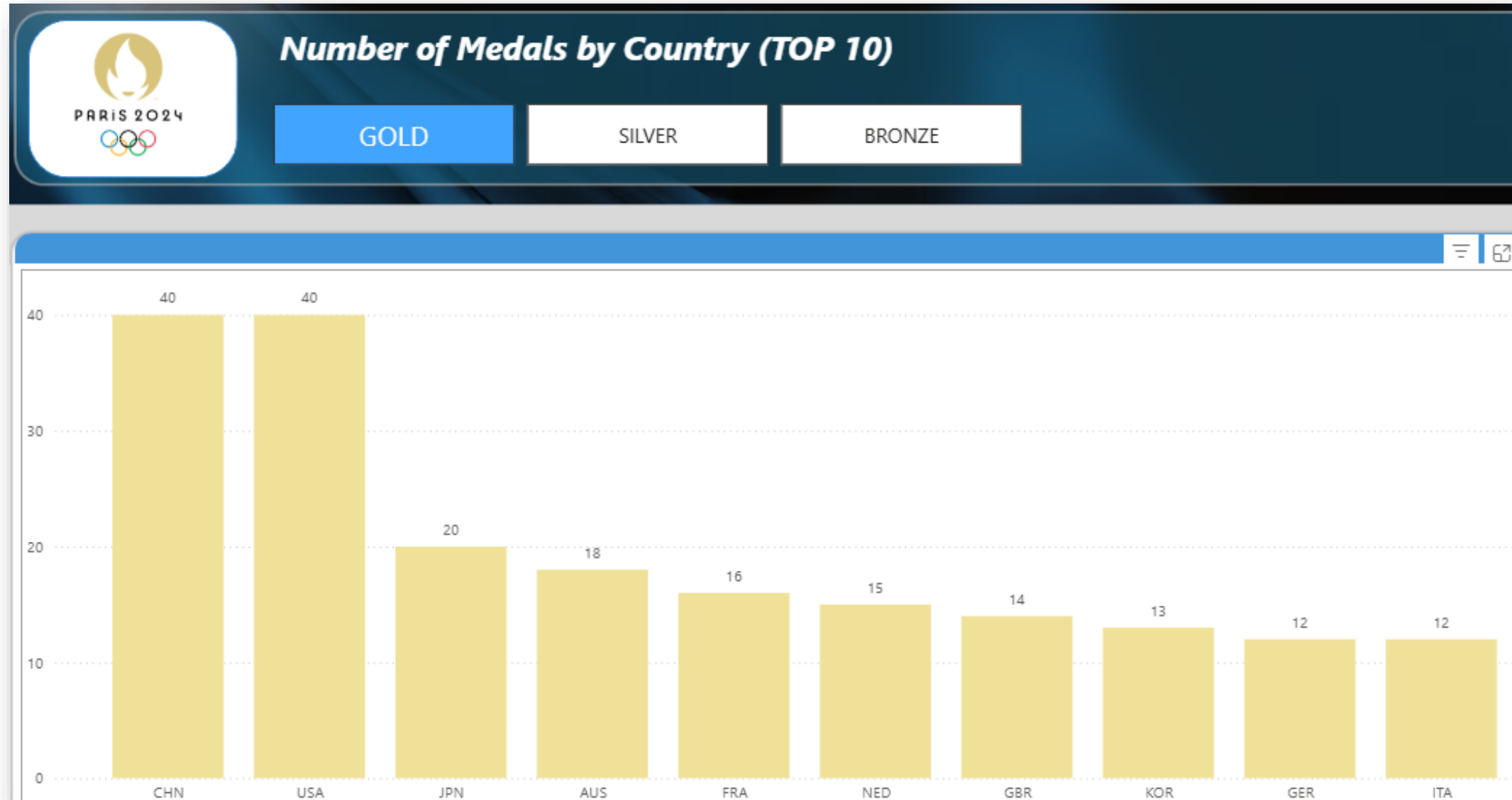
Insert 3 column charts, one for each measure created with NOCS[code] on the X axis. For each chart , go to Format > Columns > Color and choose a color matching the medal type.

Activate the bookmarks pane (View > bookmarks) then activate the selection pane and only show one visual on the page (hide the others in the selection pane) then save this snapshot as a bookmark. Repeat this process, activating one visual at a time. Create a bookmark group and regroup the 3 bookmarks there.

Insert a bookmarks navigator in the page (Ctrl + click the bookmarks). Make sure that the bookmark navigator only points to the group created earlier (Format > Bookmarks > Bookmark).

Add a scrim in the background and the official logo as an image.

Using bookmarks





FIELD PARAMETER



Power BI
Desktop

Using a field parameter

A **field parameter** in Power BI is a dynamic table that lets users **switch between different fields or measures** in visuals — without needing separate visuals or complex DAX.

Exercise:

On a new page, go to Modeling > New parameter > Fields parameter. Name it 'NB medals slicer values' and select the measures as shown on the screenshot.

This creates a new table containing links to the DAX measures and an associated slicer in the active page.

The screenshot shows the 'Parameters' pane in Power BI. The 'What will your variable adjust?' dropdown is set to 'Fields'. The 'Name' field contains 'NB Medals slicer values'. Under 'Add and reorder fields', a list of measures is shown: 'NB GOLD', 'NB SILVER', 'NB BRONZE', and 'NB Medals', each with a close button. At the bottom, the checkbox 'Add slicer to this page' is checked. On the right, the 'Fields' pane shows a search bar and a list of measures with checkboxes: 'NB BRONZE' (checked), 'NB days' (unchecked), 'NB days > AVG' (unchecked), 'NB GOLD' (checked), 'NB Medallists' (unchecked), 'NB Medals' (checked), 'NB Men' (unchecked), 'NB SILVER' (checked), 'NB women' (unchecked), 'Running Total Medals' (unchecked), and 'Variation NB medals-AVG' (unchecked). A 'day calc for medals' link is also visible. A 'Create' button is at the bottom right.

Parameters

Add parameters to visuals and DAX expressions so people can use slicers to adjust the inputs and different outcomes. [Learn more](#)

What will your variable adjust?

Fields

Name

NB Medals slicer values

Add and reorder fields

NB GOLD	×
NB SILVER	×
NB BRONZE	×
NB Medals	×

☒ Add slicer to this page

Fields

Search

- ☒ NB BRONZE
- ☐ NB days
- ☐ NB days > AVG
- ☒ NB GOLD
- ☐ NB Medallists
- ☒ NB Medals
- ☐ NB Men
- ☒ NB SILVER
- ☐ NB women
- ☐ Running Total Medals
- ☐ Variation NB medals-AVG

> day calc for medals

Create

Using a field parameter

Insert a column chart and drag the 'NB Medals slicer values[NB Medals slicer values]' to the Y-axis well and NOCS[code] to the X-axis well.

Create a DAX measure that will conditionally format the color of the columns according to the type of medal selected in the slicer as follows:

Color for sliced measure =

```
SWITCH(SELECTEDVALUE('NB medals slicer values'[NB medals slicer values Fields]),  
    "01.Measures'[NB GOLD]", "#A38600",  
    "01.Measures'[NB SILVER]", "#CCCCCC",  
    "01.Measures'[NB BRONZE]", "#F0E199",  
    "#0000AA")
```

Assign this measure to the conditional formatting of the column chart: go to Format > Columns > Color > fx and set the style as 'Field value' Then select the measure created above in 'What field should we base this on ?'

Color - Categories

Format style

Field value

What field should we base this on?

Color for sliced measure

Displaying various measures in one visual





Visual DAX



Power BI
Desktop



Adding calculation with visual DAX

- In a new page, add a table visual and drag [NB medals slicer values] in the Values well and [Date] in the rows.
- Insert a slicer based on [NB medals slicer values].
- Click on Home > Insert a new visual calculation
- In the new window that opens, select 'Running sum' by clicking the fx button then fill the formula as follows: Total to date = `RUNNINGSUM([NB GOLD])`.
- Add the `FORMAT()` function to display the running total without decimals:
Total to date = `FORMAT(RUNNINGSUM([NB GOLD]),"0")`
- Click again on fx and select "percent of grand total" to create another calculation and complete the DAX code as follows:
% of total = `FORMAT(DIVIDE([NB GOLD], COLLAPSEALL([NB GOLD], ROWS)), "0%")`

Adding calculation with visual DAX

- Click again on fx and select 'Versus previous' to create another calculation and complete the DAX code as follows:

Daily change =

```
VAR __diff = [NB GOLD] - PREVIOUS([NB GOLD])
```

```
VAR __result = IF(__diff < 0,  
    __diff,  
    "+" & __diff  
)
```

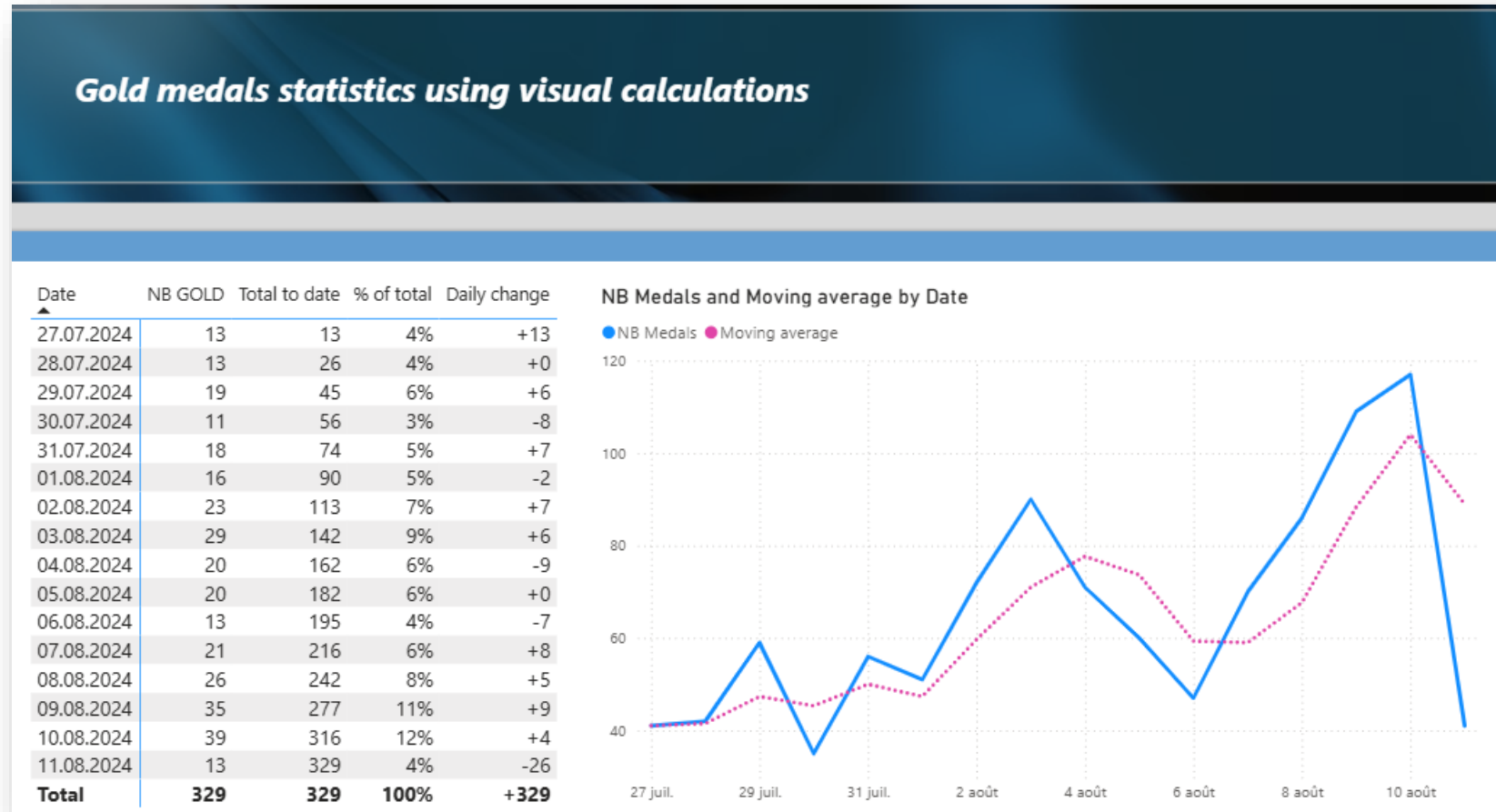
```
RETURN __result
```

Date	NB GOLD	Total to date	% of total	Daily change
27.07.2024	13	13	4%	+13
28.07.2024	13	26	4%	+0
29.07.2024	19	45	6%	+6
30.07.2024	11	56	3%	-8
31.07.2024	18	74	5%	+7
01.08.2024	16	90	5%	-2
02.08.2024	23	113	7%	+7
03.08.2024	29	142	9%	+6

Adding calculation with visual DAX

- Insert another visual calculation and choose moving average then complete DAX code as follows:

Moving average = `MOVINGAVERAGE([NB Medals], 3)`





Calculation groups



Power BI

Desktop

Using calculation groups

☐ NB BRONZE

☐ NB SILVER

☒ NB GOLD

☐ TOTAL

Statistics using calculation groups and field parameters

	Date	Current day	D-1	D-2	Var D-1	AVG 3 days	% of total	Running sum
Current day	27.07.2024	13			100,0%	4	4,0%	13
D-1	28.07.2024	13	13		0,0%	8	4,0%	26
D-2	29.07.2024	19	13	13	31,6%	15	5,8%	45
Var D-1	30.07.2024	11	19	13	-72,7%	14	3,3%	56
AVG 3 days	31.07.2024	18	11	19	38,9%	16	5,5%	74
% of total	01.08.2024	16	18	11	-12,5%	15	4,9%	90
Running sum	02.08.2024	23	16	18	30,4%	19	7,0%	113
	03.08.2024	29	23	16	20,7%	22	8,8%	142
	04.08.2024	20	29	23	-45,0%	24	6,1%	162
	05.08.2024	20	20	29	0,0%	23	6,1%	182
	06.08.2024	13	20	20	-53,8%	17	4,0%	195
	07.08.2024	21	13	20	38,1%	18	6,4%	216
	08.08.2024	26	21	13	19,2%	20	7,9%	242
	09.08.2024	35	26	21	25,7%	27	10,6%	277
	10.08.2024	39	35	26	10,3%	33	11,9%	316
	11.08.2024	13	39	35	-200,0%	29	4,0%	329
	Total	329	316	277	4,0%	307	100,0%	329

Calculation group: Set up

- Go to Model view and in the Data pane, select the 'Model' tab and right click on 'Calculation groups' > New calculation group.
- Rename the default calculation item to 'Current day'.
- Right click on 'Calculation item' > New calculation item and insert the following DAX code:

D-1 =

```
VAR __prevday = CALCULATE(SELECTEDMEASURE(), DATEADD('CALENDAR'[Date],-1,DAY))
VAR __result = IF(NOT ISBLANK(__prevday),__prevday,"")
RETURN __result
```

- Repeat with:

D-2 =

```
VAR __minus2days = CALCULATE(SELECTEDMEASURE(), DATEADD('CALENDAR'[Date],-2,DAY))
VAR __result = IF(NOT ISBLANK(__minus2days),__minus2days,"")
RETURN __result
```

- Add another calculation item:

Var D-1 =

```
VAR __previous = CALCULATE(SELECTEDMEASURE(),DATEADD('CALENDAR'[Date],-1,DAY))
VAR __diff = SELECTEDMEASURE() - __previous
VAR __result = DIVIDE(__diff, SELECTEDMEASURE())
RETURN
IF( ISBLANK(SELECTEDMEASURE()) || ISBLANK(__previous),
    BLANK(),
    FORMAT( __result,"0.0%")
)
```


Calculation group: Set up

- Add the 3 days average calculation item :

AVG 3 days =

```
VAR __currday = SELECTEDMEASURE()  
VAR __prevday = CALCULATE(SELECTEDMEASURE(),DATEADD('CALENDAR'[Date],-1,DAY))  
VAR __minus2day = CALCULATE(SELECTEDMEASURE(),DATEADD('CALENDAR'[Date],-2,DAY))  
VAR __result = ROUNDDOWN(DIVIDE(__currday+__prevday+__minus2day,3),0)  
return __result
```

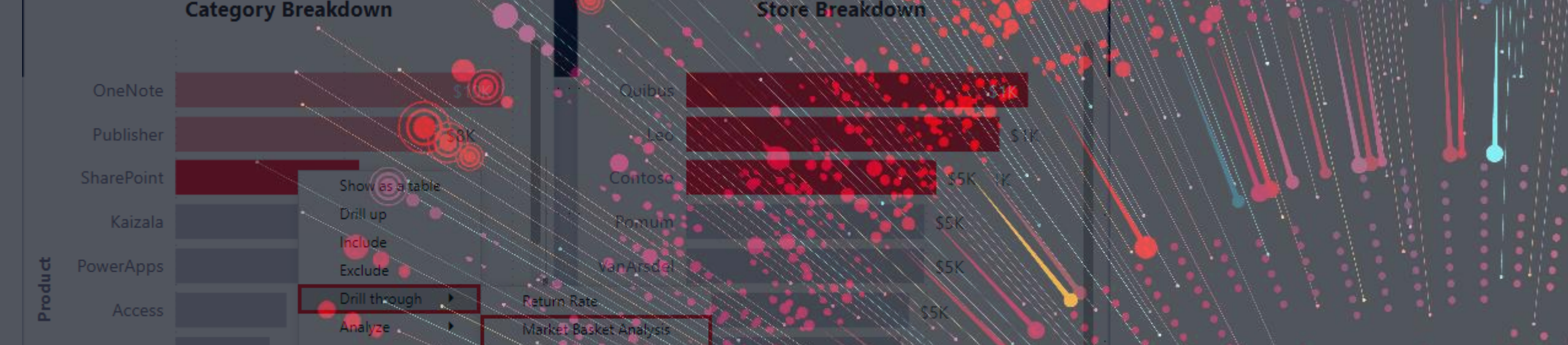
- Create the '% of total' calculation item:

% of total =

```
VAR __pct =  
    DIVIDE(  
        SELECTEDMEASURE(),  
        CALCULATE( SELECTEDMEASURE(), ALL(CALENDAR[Date]))  
    )  
VAR __result = FORMAT(__pct,"0.0%")  
RETURN __result
```

Calculation group: Set up

- Rename the calculation group 'day calc for medals'
- Go back to report view, create a new page and insert a table visual with Calendar[Date] in the Rows and the calculation group column in the Columns. Drag the field parameter created earlier [NB medals slicer value] in the values.
- Insert a button slicer with the calculation group column again. The slicer now shows all calculation items available and you can select one calculation or more to be displayed. The medals slicer allows you to view the different calculations over the selected measure.



DRILLTHROUGH PAGE

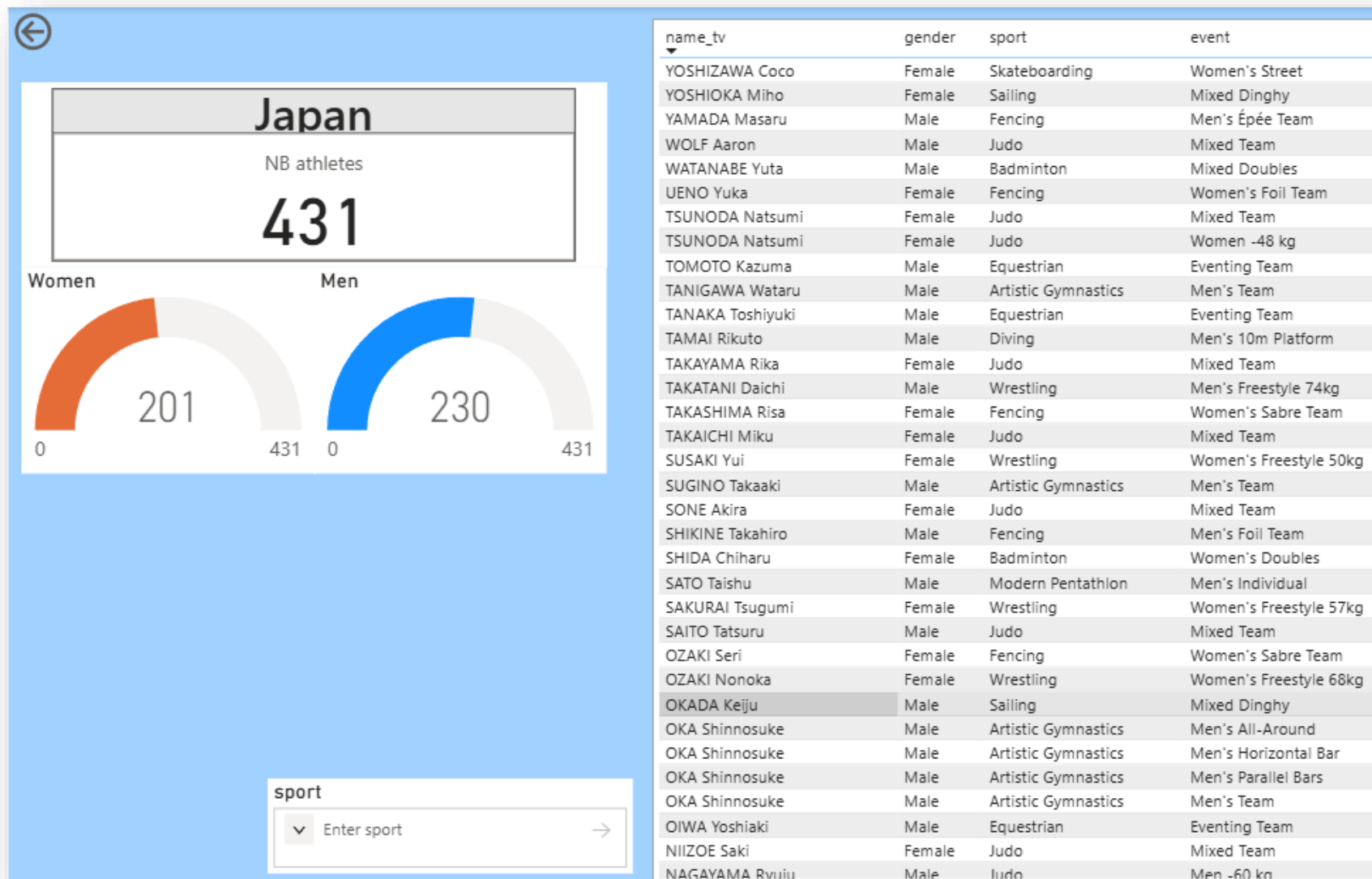


Power BI
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Drillthrough page

- Create a new page and in the page format section, select 'drillthrough' under Page Type then drag NOCS[country_long], NOCS[code] and NOCS[nocID] in the 'Drill through from' well.
- Add a card visual with [NB Athletes] in the Value well and [country_long] in the Categories.
- Add 2 gauges visuals with [NB Men] & [NB Women] and [NB Athletes] as the maximum value .
- Add a table visual and drag the following fields in the 'Columns' well : Paris2024_ATHLETES[name_tv], Paris2024_ATHLETES[gender], Paris2024_EVENTS[sport], Paris2024_EVENTS[event].
- Add an input (text) slicer to the page and drag Paris2024_EVENTS[sport] in the Field well. You can now search for specific values in the table.
- Wherever the country name or code is used in any visual of the report, the drillthrough option will now be enabled and will point to the current page.
- Change the background color of the page to make it stand out.

Drillthrough page





TOOLTIP PAGE



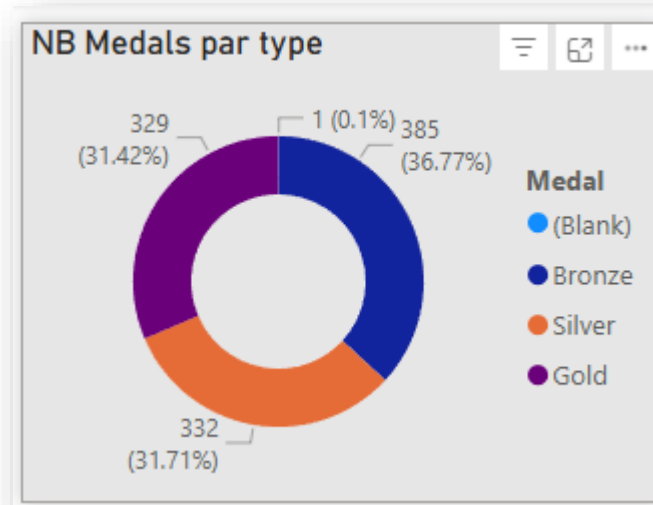
Power BI

Desktop



Tooltip page

- Create a new page and in the page format section, select 'drillthrough' under Page information > Page Type then drag NOCS[country_long], NOCS[code] and NOCS[nocID] in the 'Drill through from' well.
- Add a donut visual with [NB Medals] in the Value well and Medal Sort Table[Medal] in the Legend.
- Change the background color of the page to create contrast.
- This tooltip page will not display every time you hover the mouse over a visual that contains one of the 3 fields from the NOCS table.





Introduction to DAX QUERY VIEW



Power BI

Desktop

DAX QUERY VIEW

What is DAX Query View?

- A dedicated environment in Power BI Desktop to write, run, and analyze DAX queries
- Based on DAX Query Language (EVALUATE, VAR, RETURN)
- Allows querying the semantic model without creating visuals

What It Is NOT:

- ❌ It does not replace visuals
- ❌ It does not store results in the model
- ❌ End users never see it — developer tool only

Main Goal: Explore, test, and debug DAX logic directly against the data model

Create your first query

- Insert the following code in a new query and click 'Run':

```
EVALUATE  
SUMMARIZE(Paris2024_MEDALS, 'CALENDAR'[Date],  
"NB", [NB Medals])
```

- Take the code to the next level by adding this code under the previous one:

```
EVALUATE  
ADDCOLUMNS(  
    SUMMARIZE(Paris2024_MEDALS, 'CALENDAR'[Date]),  
    "@NB", [NB Medals],  
    "@NB GOLD", [NB GOLD],  
    "@pctgold", FORMAT(DIVIDE([NB GOLD], [NB Medals]), "0.0%"),  
    "@AVG", [AVG medals]  
)
```


Using DAX query view

- View the table used in the combo chart measure:

```
EVALUATE  
  
VAR __vt1 =  
    ADDCOLUMNS(  
        SUMMARIZE(Paris2024_MEDALS, 'CALENDAR'[Date]),  
        "@NB", [NB Medals]  
    )  
  
VAR __avg = [AVG medals]  
  
VAR __vfilt1 = FILTER(__vt1, [@NB] > __avg)  
  
RETURN __vfilt1
```

